Exercises : Chapter 3 and 4

Course: Business Intelligence (BI)
Level: 4th Year Artificial Intelligence Engineering

Exercise: Furniture Sales

Part 1: Designing a Data Warehouse

A furniture sales company wants to create a data warehouse. Consider the following information:

- The data warehouse should allow analysis with respect to the furniture sold, customers, and time.
- A piece of furniture is associated with a type (chair, table, display cabinet, etc.), a category (kitchen, living room, bathroom, etc.), and a material (wood, marble, etc.).
- A customer should be associated with a location (city, region, and country).
- The company is interested in the number of sales, revenue, and discounts on completed sales.

Question 1. Design a multidimensional model for the data warehouse based on the requirements above. You must explicitly specify the **grain** of the analysis, the **dimensions**, and the **facts**.

Question 2. Develop a star schema for a data warehouse corresponding to the above requirements.

Question 3. Develop a snowflake schema for a data warehouse corresponding to the above requirements.

Question 4. The company management decides that storing every single transaction is too expensive. They now only want to analyze the total revenue per Month and per Furniture Category. Describe how this change in **granularity** affects the Fact Table and the Dimensions.

Part 2: Data Analysis and KPI Calculation

Assuming the Data Warehouse is populated with sales data, the management wants to build a Dashboard to monitor performance.

Based on the Star Schema designed in Question 2 Part 1, write the SQL queries to calculate the following KPIs:

1. **Category Performance:** Calculate the Total Revenue generated by each *Furniture Category* in 2024.
2. **Customer Ranking:** Identify the top 3 customers in each region based on their Total Spending. *(Hint: You need to calculate total spending per customer first, then rank the results).*
3. **Trend Analysis:** Calculate the Monthly Sales Growth (Difference vs Previous Month) for the year 2024. *(Hint: You need to aggregate sales by month first, then compare with the previous month).*

4. **Cumulative Progress:** Calculate the Year-to-Date (YTD) Revenue for the "Kitchen" category to show how revenue accumulates day by day in 2024.

5. **Price Comparison:** Display the revenue of each individual sale alongside the **Average Revenue** of its Category (for comparison).

Important Instructions:

- For **queries (2 to 5)**, you are **required** to use **Window Functions** (See Chapter 4).
- For complex queries (like Question 2 and 3), use the `WITH` clause to create a temporary "helper table" to simplify your code before applying the Window Function.

```
WITH Temporary_Table_Name AS (
    SELECT ... -- Calculate intermediate results (e.g., Sums)
    FROM ...
    GROUP BY ...
)
SELECT ...
FROM Temporary_Table_Name
WHERE ...; -- Perform final ranking or analysis
```